

Recovering Sparse Neural Connectivity from Partial Measurements

A Covariance-Based Approach with Granger-Causality Refinement

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github.com/qsimeon/sparse_matrix_recovery

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Problem: Partial Observability

Goal: Recover $N \times N$ connectivity W from **partial measurements** across sessions.

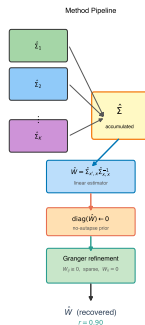
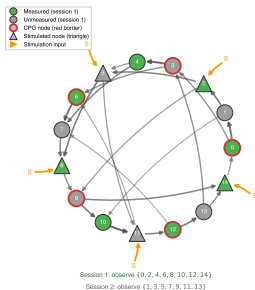
Dynamical model:

$$x_{t+1} = W\phi(x_t) + b_t$$

Session k observes $\mathcal{M}_k \subset \{1, \dots, N\}$, stimulates $\mathcal{S}_k$.

Key insight: Pairwise covariance can be **accumulated across sessions** whenever i, j are co-observed.

C. elegans, zebrafish, organoids



Method: From Dynamics to Estimator

Step 1 — Covariance accumulation over $K=50$ sessions:

$$\hat{\Sigma}_{x_{t+1}, x_t}(i, j) = \frac{1}{n_{ij}} \sum_{k: i, j \in \mathcal{M}_k} \hat{\Sigma}^{(k)}(i, j)$$

Step 2 — Linear estimator. Multiply $x_{t+1} = W\phi(x_t) + b_t$ by $x_t^\top$, take expectations:

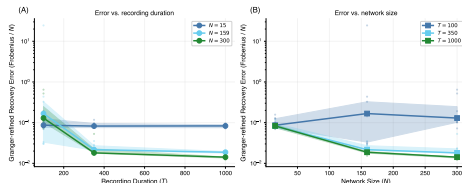
$$\Sigma_{x_{t+1}, x_t} = W \Sigma_{\phi(x), x} + \Sigma_{b, x} \xrightarrow{\phi(x) \approx x} \boxed{\hat{W} = \hat{\Sigma}_{x_{t+1}, x_t} \hat{\Sigma}_{x_t, x_t}^{-1}}$$

with diagonal zeroed ($W_{ii}=0$, no autapses) and non-negativity ($W_{ij} \geq 0$).

Step 3 — Granger-causality refinement: Projected gradient descent enforcing sparsity zeros where contemporaneous covariance exceeds lagged.

Identifiability: $K \geq \log(N^2/\delta)/p^2$ sessions suffice for all-pairs coverage with probability $\geq 1-\delta$.

Result: $r = 0.90$ at 66% Coverage



Error decreases with T and N .

Correlation

$r = 0.90$

Over same-prior

31% improvement

Best error

0.014 ($N=300, T=1000$)

Coverage

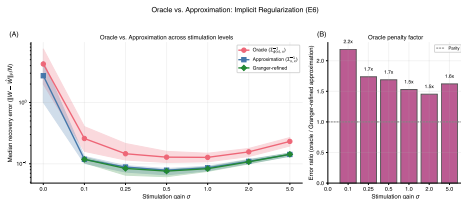
66% / session

Sessions

$K=50$

Granger refinement: precision
0.30→0.40, recall 0.97 — nearly all true
edges preserved.

Surprise: The “Wrong” Model Wins



Oracle with known $\phi = \tanh$ is
1.5–2.7× worse.

We *deliberately* ignore the known nonlinearity.

Why? Stein–Price identity gives $\Sigma_{\phi(x),x} = D\Sigma_{x,x}$ where $D = \text{diag}(\mathbb{E}[\text{sech}^2(x_i)]), 0 < d_i < 1$.

- ▶ Oracle inverts $D\Sigma$ — **ill-conditioned**
- ▶ Linear inverts Σ — biased but **better-conditioned**

⇒ **James–Stein shrinkage** in action.
No need to characterize the nonlinearity.

The Real Bottleneck: Intrinsic Dynamics

The estimation error decomposes into two terms:

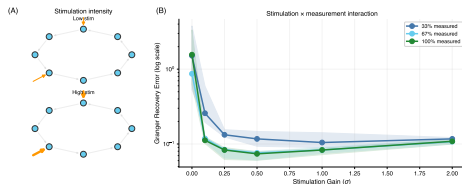
$$\underbrace{\hat{W} - W}_{\text{total}} = \underbrace{W(D - I)}_{e_1: \text{model mismatch}} + \underbrace{\Sigma_{b,x} \Sigma_{x,x}^{-1}}_{e_2: \text{CPG correlation}}$$

Error Source	$\ E\ _F/N$	Relative
e_1 : Model mismatch (linear $\approx$ tanh)	0.080	1×
e_2 : CPG–state correlation	0.225	2.8×

Better recovery requires **modeling autonomous dynamics**,
not refining the nonlinear approximation.

Practical Guidelines for Experimentalists

1. **Stimulate** moderately ($\sigma \approx 0.5\text{--}1.0$) on $\sim 33\%$ of neurons
2. **Cover** $\geq 50\%$ neurons per session
3. **Record** $K \geq 50$ sessions with random masks
4. **Use the linear estimator** — simpler *and* better
5. **Next:** Model CPG dynamics ($\sim 3\times$ dominant error)



The **control–estimation tradeoff**:
stimulation aids identifiability
but disrupts the signal.

**Covariance accumulation turns partial observation
from a limitation into a feature.**

Paper & Code github.com/qsimeon/sparse_matrix_recovery

Contact qsimeon@mit.edu

Thank you!

Questions welcome at the poster session.